MOBILE ROBOTICS LAB 8

Summary of Customizations

I have designed and implemented a custom mobile robot by extending the basic URDF structure provided in the lab manual. My customizations focus on incorporating diverse geometries and experimenting with different joint types to enhance the robot's functionality and visual appearance.

1. **Link and Geometry Additions:**
   - **Chassis:** I created a rectangular body using a **Box** geometry as the primary base link.
   - **Wheels:** I added two cylindrical wheels to the sides of the chassis.
   - **Head Unit:** I implemented a **Sphere** on top of the chassis to serve as a specialized sensor/head unit.
   - **Antenna:** I added a decorative and functional antenna using a long, thin **Cylinder** attached to the head.
2. **Joint Implementation:**
   - **Continuous Joints:** I used continuous joints for the left and right wheels, allowing them to rotate indefinitely for driving simulation.
   - **Revolute Joint:** I added a revolute "neck" joint connecting the chassis to the head. I set specific joint limits to allow the head to rotate 90 degrees in both directions, which I verified using the Joint State Publisher sliders in RViz.
   - **Fixed Joint:** I used a fixed joint to securely mount the antenna to the head, ensuring it stays in place during rotation.
3. **Visualization Improvements:**
   - I assigned distinct colors to each component (Blue for the body, Black for the wheels, White for the head, and Red for the antenna) using RGBA material tags to ensure the robot is easily distinguishable in the RViz environment.
   - I successfully configured the Fixed Frame to `base_link` in RViz to view the robot model and its TF tree accurately.